

Global Nickel Short-Term Outlook

January 2026



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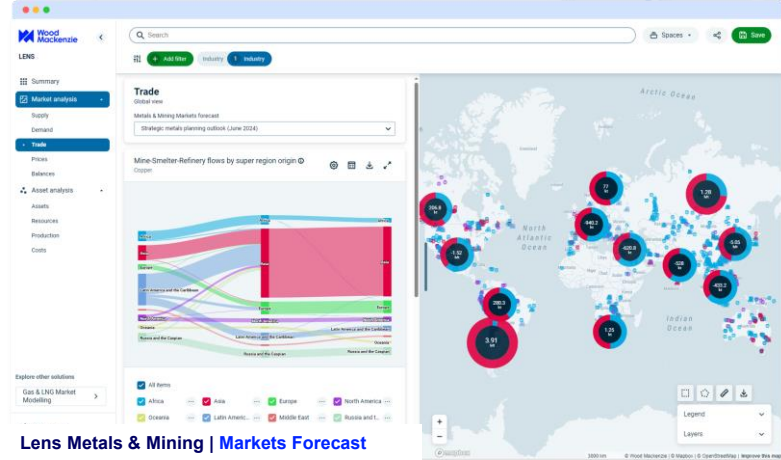
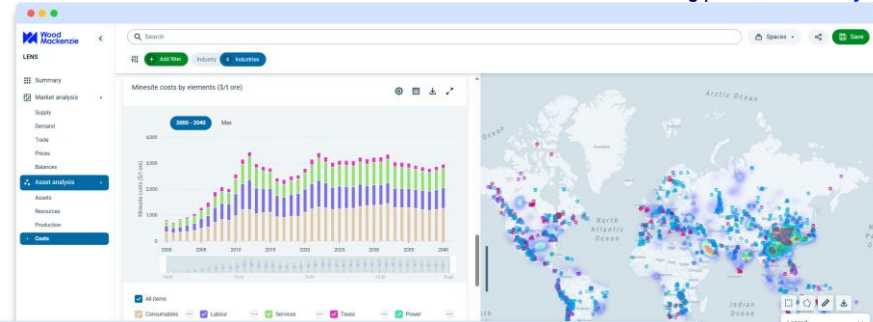
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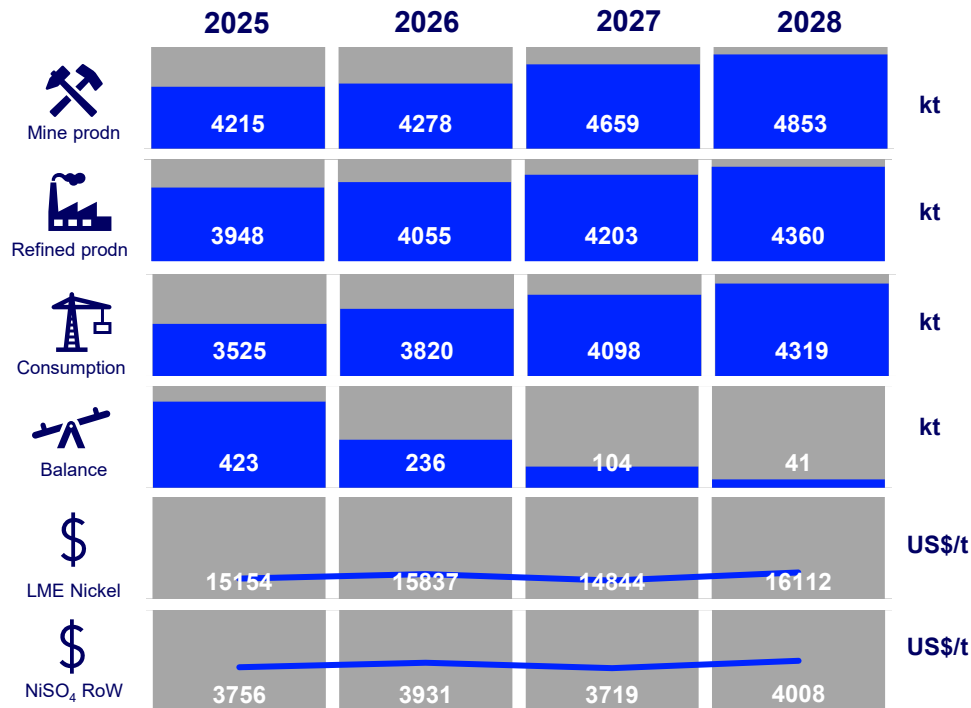
Executive summary: Dr. Copper's and Mr. Nickel's wild rides

Nickel hit a peak price of US\$18,628/t (US\$8.45/lb) late this month, up by almost 32% on the eight-month low of US\$14,118/t on 16 December. Copper's *average* cash price through to 27 January rose by 10.4% to a record US\$13,035/t (US\$5.90/lb) from December. Aluminium moved beyond the US\$3,000/t mark for the first time since April 2022, lithium has moved up sharply again, tin, gold, silver, PGMs, all up at the same time. The raw materials and resources panic is on, sustained by a range of views from a short-term supply constraint fear to a longer-term demand push due to militaristic self-protectionism in an increasingly insecure world.

Nickel prices are benefitting from this wider commodities availability concern. But the real driver for our metal remains the headlines – believable or otherwise – emanating from Indonesia regarding the timely issuance of mining permits. The confusion in the market over how much miners will be able to extract and process has been going on since late-September. And now that the government has extended its own deadline out to end-March before issuing such permits, the chaos will live on.

Imagine a world in which this single issue goes away for nickel, and so will (largely) the prevailing high prices. This is the view we are taking still in light of no “real” news. And if/when the headline permitted tonnes figure comes through and proper analysis can be done, only then will we see the real contribution to nickel prices of the noise boosting all other metals around us right now.

Nickel key forecasts



Source: Wood Mackenzie

Nickel balances by type

Class I – There is very little change on our balance split since December's report. Last year's Class I excess will continue through our forecast period but at a declining rate. Most new cathode production from China, and now from two sites in Indonesia, is being put into LME depots.

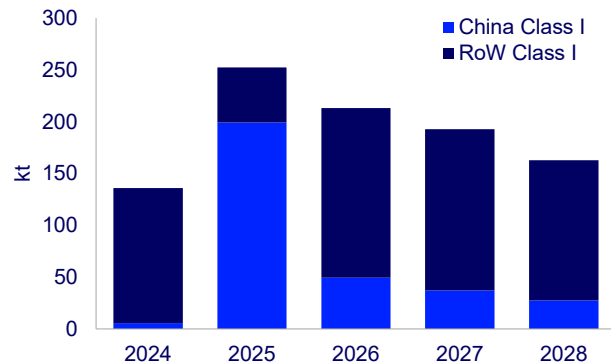
Class II – That Class II markets in China appear to be heavily oversupplied in 2025 is due to an apparent overshoot in supply caused by the substantial ramp ups in NPI at Weda Bay, Halmahera Jaya and Huabao. A reassessment of China's stainless product mix and scrap use in February may lead to changes.

Nickel sulphate – The global sulphate market appears broadly balanced, with China having spare material while the rest of the world is short. Easing any supply tightness in China by producing more sulphate would at the same time reduce Class I metal output.

	2022	2023	2024	2025	2026	2027	2028
Class I							
China Class I	20	7	6	199	49	37	28
RoW Class I	-43	11	130	51	175	158	136
Total Class I	-23	19	136	251	224	195	164
Class II							
China Class II	163	202	-2	184	13	-85	-146
RoW Class II	16	11	75	-10	0	0	0
Total Class II	179	213	73	174	13	-85	-146
Nickel sulphate							
China sulphate	-7	-4	31	31	59	23	-11
RoW sulphate	-24	-14	-12	-32	-60	-29	34
Total sulphate	-32	-19	19	-2	-1	-6	22
Total nickel	125	213	228	423	236	104	41

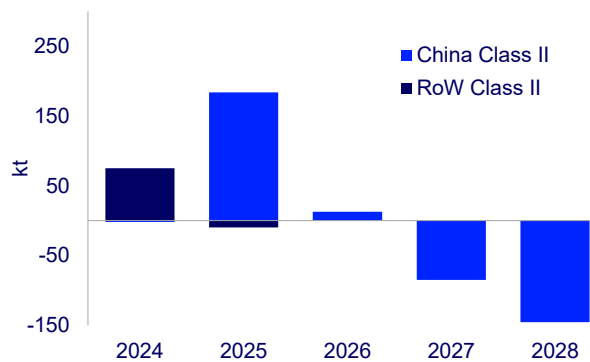
Source: Wood Mackenzie

Class I nickel balance



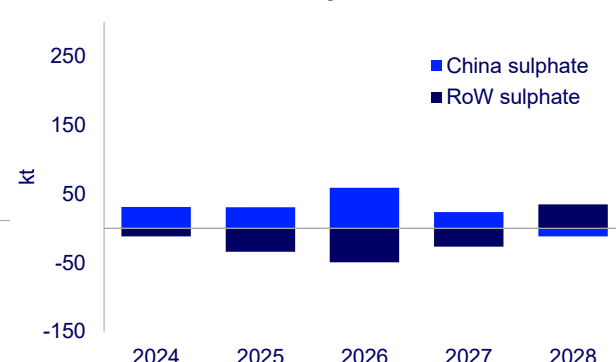
Source: Wood Mackenzie

Class II nickel balance



Source: Wood Mackenzie

Nickel sulphate balance



Source: Wood Mackenzie

Prices

Nickel prices rally from mid-December - 1

Prices rise US\$4,500/t in three weeks, yet LME stocks also increase – up 20 kt in a single day

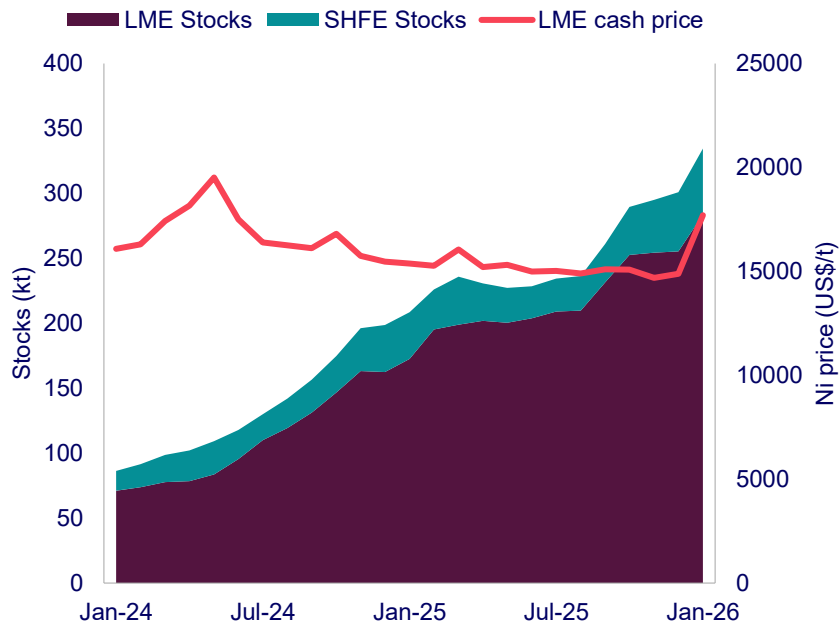
Nickel prices have always been volatile and with so little movement away from US\$15,000/t through most of 2025 a sudden change was perhaps to be expected. When it came the rally was faster and scaled greater heights than expected, especially since nothing had changed the presiding oversupply in the market.

Nickel prices turned after reaching a low of US\$14,117/t (US\$6.40/lb, cash) on 16 December. The rebound accelerated on 23 December and again a week later to US\$16,470/t (US\$7.45/lb) before a final surge starting on 6 January took the price up to US\$18,450/t (US\$8.35/lb) on 7 January, its highest value since mid-2024. A sharp drop the following day was followed by two weeks of relative stasis between US\$17,600/t and US\$18,000/t (US\$8.00/lb), before another surge to the month's peak of US\$18,627/t (US\$8.45/lb) at the close on 23 January.

Yet exchange stocks went the other way. SHFE stock increased by 5 kt through January while LME inventory increased by 31 kt. Two thirds of the LME total - 20 kt – entered warehouses on a single day, 6 January, with a further 8 kt arriving on 8 January. The receiving warehouses were mostly in Kaohsiung in Taiwan. Total exchange nickel stocks stood at 329 kt on 27 January.

Technically, the bulls had taken control with new long positions and short covering dominating the period from 16 December to 23 January. However, since then shorts have come into play, as if in acknowledgement of the fact that the market is still oversupplied. In that sense nothing has changed from late last year. Does this signal the end of the bull run and a retreat to mid-December's starting position?

Exchange nickel stocks and prices



Source: LME; SHFE; Wood Mackenzie

Nickel prices rally from mid-December - 2

Rally fueled by hearsay surrounding Indonesian mining quotas and the impact on downstream production

The Explanation: The rally in nickel prices was based on speculation over the result and impact of the Indonesian government's setting of nickel ore mining quotas (RKAB) for 2026. Since October 2025 the *expectation* has been that constraints on Indonesian mining were necessary to slow smelter and refinery production to rein in the oversupply. At the very least, the switch from three-year mining permits back to one-year was expected to cause some disruption to ore supply that would push nickel prices higher in Q1 2026. So, the market has been primed for a rally since then.

We have already reviewed in detail the impact on nickel prices of the early-year events in Indonesia in the insight: "[Nickel: 5 things to look for in 2026](#)". The key announcement was that RKAB quotas – quotas not production - might be cut to 250 Mt, compared with 379 Mt in 2025. Not surprisingly such a reduction has substantial implications for production in 2026 and while the cut was no more than a proposal – and remains so – this gave the nickel price upward momentum.

Then on 2 January we learned that PT Vale had not received its 2026 RKAB and had to stop mining – a signal to the market that perhaps other companies would also be forced to suspend mine operations. Prices jumped - only to then realise while Vale's permit had ended in December 2025, all other companies had pre-existing permits for 2026 that only needed re-approval, and that they would be allowed to continue mining using that permit until the end of Q1 2026 while their re-approvals were granted.

Also at this point, all metals commodities prices went up. Whether this was tied to the US's "Fracas in Caracas" or not, this too added further uplift to nickel.

Profit-taking on 8 January coincided with the warranting of 8 kt nickel in LME warehouses and prices fell back. In the following days it became clearer what the proposed cut in *quotas* might mean for ore *production* and, as we point out in the insight "[Will Indonesia keep feeding its nickel giants](#)", if the quota of 250 Mt is confirmed then there might not be enough ore to satisfy the feed volumes needed by the smelters to meet our 2026 forecast. While this is significant and reasonably enhances the rationale for higher prices it says nothing about the ability of ore stocks and ore imports from Philippines to fill the shortfall.

Uncertainty prevailed and prices settled at US\$17,600-18,000/t only to be temporarily disrupted, seemingly, by news of the sinking of the MV Devon Bay cargo ship carrying 55 kwmt nickel ore from Zamboanga Nickel Corp in the Philippines to Indonesia overnight 22-23 January. This sent the nickel price to its highest level for the month of US\$18,627/t, with intra-day values even higher, on fears that this loss of ore would further tighten the supply of ore to NPI smelters. However, tragic though the sinking was, the lost cargo was small in nickel terms, and prices have since come off.

The Outlook: So, what does this all mean for nickel prices in the months ahead?

Clearly, a US\$4,500/t (US\$2.00/lb) increase in the nickel prices in only three weeks means that our forecast for 2026 has increased since the December Short-Term Outlook. However, with so much currently depending on Indonesia's RKABs and whether this is transformational for production, the lack of decision means it is difficult see far ahead. In fact, given Indonesia's allusion to adjusting quotas so they align with smelter production plans, it is possible that quotas could be cut from 2025's level, yet still enable our smelter and refinery production forecast to be achieved.

Nickel prices rally from mid-December - 3

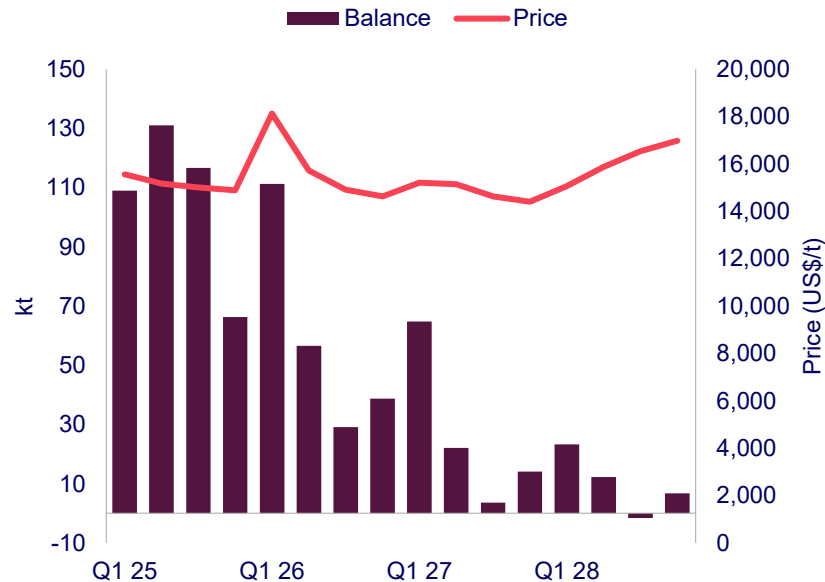
Elevated prices to remain through Q1 before retreating, unless there is clarity in Indonesia

Initial indications were that a decision on quotas would be made by early January 2026. However, as companies are allowed to continue mining until the end of March 2026, while they await RKAB approval, there is now no rush to make that decision so hastily.

In the absence of a decision, we cannot make any adjustment to our supply outlook, so our nickel market remains oversupplied in 2026, and our default position is still that prices will decline through the year. Ongoing uncertainty and anticipation will keep nickel prices buoyant throughout Q1, averaging US\$18,130/t (US\$8.20/lb), but thereafter retreat towards US\$15,000/t to reflect the surplus that we continue to show. This leaves the forecast average nickel price for 2026 at US\$15,850/t (US\$7.20/lb) but this will change if we learn that Indonesian quotas are confirmed to the detriment to our forecast of downstream nickel production.

The market surplus continues in 2027 with prices very similar to 2025, at US\$14,500-15,500/t (US\$6.80/lb). However, supply starts to tighten in H2 2027 and this extends in 2028, when the market surplus shrinks to only 41kt, which is essentially balanced. This year will see prices turning the corner, averaging around US\$16,100/t (US\$7.30/lb) with spot highs late in the year of US\$17,000-18000/t (US\$7.95/lb), on their way to higher values in later years, when supply falls behind demand.

Quarterly nickel market balance and price



Source: Wood Mackenzie

SHFE goes international

More work is needed to offer real competition to the LME, but Beijing and SHFE are making their intent clear

The “internationalisation” of the Shanghai Futures Exchange (SHFE) received a long-anticipated boost this month from the Shanghai Municipal Financial Regulatory Bureau, whose policy document - “*Action Plan to Strengthen Futures-Spot Linkage and Enhance the Capacity of Non-Ferrous Metals Commodities*” - was released on 21 January. Largely confirming the main topic of discussion in the aforementioned insight “[Nickel: 5 things to look for in 2025](#)” – the Bureau’s plan outlined three “Strategic Pillars” containing 18 measures it would take to achieve its goals. The second “Pillar” is of most relevance to the conventional way nickel has been traded globally on the LME since the 1980s: “Internationalisation – opening futures and options to global investors and establishing overseas delivery mechanisms”. In more detail, the pillar intends to “...expand the high-level institutional opening of the Shanghai non-ferrous metal futures market, gradually include the varieties of non-ferrous metal product futures and options that meet the conditions into the range of varieties open to the outside world, and simultaneously improve the business details compatible with those varieties. Actively explore the innovation of cross-border settlement mechanisms, deepen the “going out” of settlement services, and establish overseas warehouses and cross-border settlement business models under the premise of controllable risk.”

Subsequently, on 23 January, the China Securities Regulatory Commission (CSRC) published a notice to implement the Bureau’s policy “...urging the relevant futures exchanges to make all necessary preparations to ensure the smooth and orderly introduction of 14 additional contracts for participation by overseas traders.” Combined with the 24 previous designated contracts, the total number of the internationalized contracts has now reached 38, further opening China’s financial market to foreign investors.

In short, the Bureau and the Commission are granting China’s commodities exchanges, such as the SHFE, INE and GFEX, breathing room to “...strengthen the influence of the “Shanghai price””. Among the additional 14 contracts to benefit from such immediate regulatory permission and support are the nickel futures and options currently on the SHFE and the lithium carbonate futures and options listed on the Guangzhou Exchange. For the sake of completion, within our base metals wheelhouse the bonded copper options contract now on the Shanghai International Energy Exchange (INE) will also be similarly opened up, but a date has not been revealed yet.

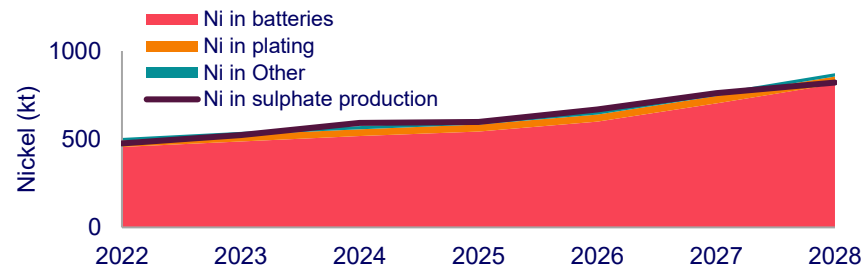
The Ni/Li contracts on the SHFE/GFEX have been available to the international trading community before, but only through two specialised routes. This much easier third route confirms China’s intent to attract base and energy metals trading activity east to where production and consumption largely take place.

Nickel sulphate supply/demand and price

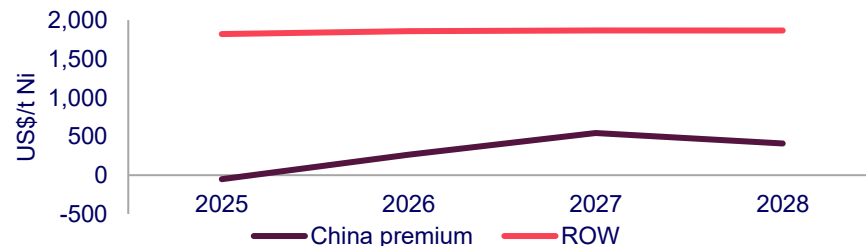
Nickel sulphate premiums diverge, with China maintaining domestic strength while Indonesia trades at discount

- China:** In January, domestic nickel sulphate premiums remained elevated, which has persisted for six months. Despite this, our 2025 annual average domestic price stands at a discount of US\$51/t nickel, while our 2026 annual domestic premium is forecast at around US\$300/t nickel, amid resilient MHP payability and stable demand from the EV sector. Additionally, as the Chinese New Year approaches (15–23 February), spot market activity picked up in January for stockpiling from pCAM producers, further underpinning the sulphate price.
- Rest of World:** Sulphate premiums are expected to sit at US\$1,800/t nickel in 2026 amid high production costs.
- Indonesia:** Customs data show that Chinese imports of Indonesian nickel sulphate still traded at a discount. The 2025 average discount could stay around US\$2,200/t nickel. Most materials remain committed under long-term supply agreements.

Global NiSO₄ supply and demand balance



Nickel sulphate premia/discount (US\$/t Ni, real)



Source: the Argus media group, Global Trade Tracker, Wood Mackenzie

Demand

Changes in forecast nickel demand this month

Chinese stainless strengthens a bit for January while Delong's Indonesian mill has been offline since September

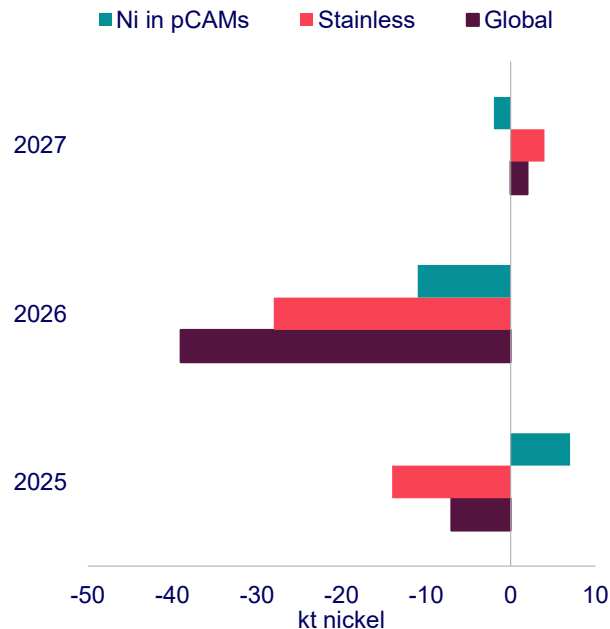
Moderate revisions were made to nickel consumption in both stainless and battery materials, almost exclusively in Asia.

China: A stronger-than-expected December added 100 kt stainless to the national total in 2025. In contrast, nickel-in-pCAM was unchanged although the outlook for 2026 and 2027 was reduced slightly. A cut in export rebates from 1 April may cause a surge in exports – and production – in Q1 2026.

Indonesia: The latest Indonesian stainless data showed that Delong's Obsidian plant has been offline since September 2025. We have pushed back a re-start to later this year and reduced the outlook by 280 kt in 2026. Start-up at CNGR's pCAM plant in Kalimantan (see [slide # 19](#)) raised our forecast of Indonesian nickel-in-pCAM demand by 6 ktpa.

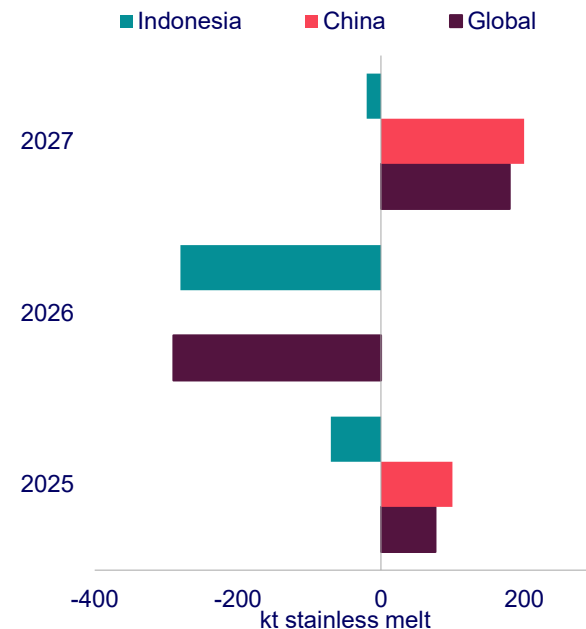
CNGR also confirmed to us that its Moroccan pCAM plant is operating so this has been reinstated with a consequent 5-10 ktpa uplift in African nickel-in-pCAM demand compared with last month.

Main changes in nickel consumption



Source: Wood Mackenzie

Main changes in stainless melt output



Source: Wood Mackenzie

Short-term outlook for global nickel consumption

China, Indonesia and India to drive growth in 2026, while Europe tries to bounce back

Global nickel consumption increased by 2.4% in 2025, to 3.53 Mt, largely driven by China where demand in both stainless and battery pCAMs was moderately better, the latter improving in H2.

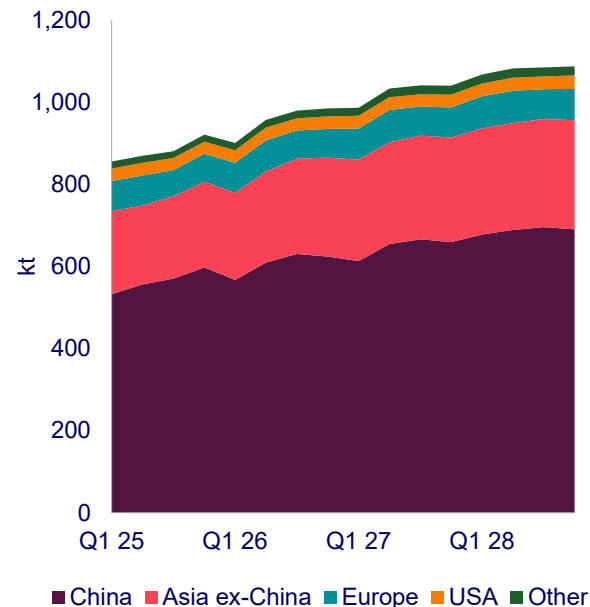
Global growth will strengthen in 2026, rising by 8.4% to 3.82 Mt. Most of this again is in China (+8%) and Indonesia (+18%) where stainless and pCAM demand will rise supported particularly by new pCAM capacity in both countries. India's stainless market is also growing.

That Europe will enjoy some growth in 2026 is reasonable after contractions in three of the past four years. While booming demand for nickel alloys will support this (and particularly in the US too), signals from the stainless market already suggest a downgrade to our European forecast might be needed in the near term.

Forecast global nickel consumption will increase by another 7% to 4.1 Mt in 2027, before slowing to 5%, to 4.3 Mt in 2028.

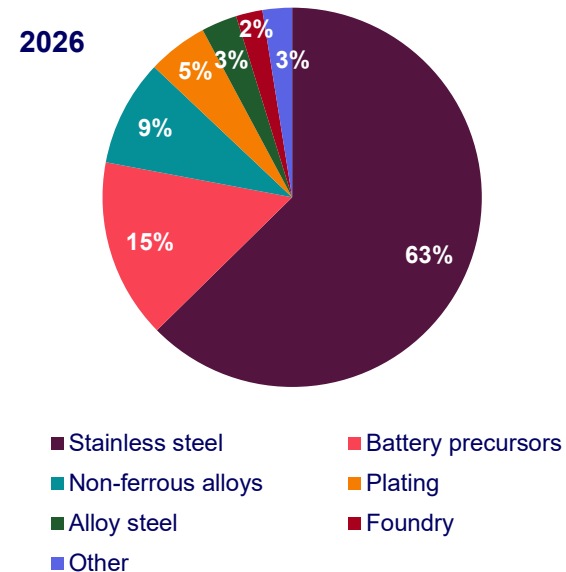
Stainless remains the principal consumer of nickel although the share of global demand going to pCAMs will increase from 15% in 2025 to 18% in 2028.

Refined nickel consumption by region



Source: Wood Mackenzie

Refined nickel consumption by first use



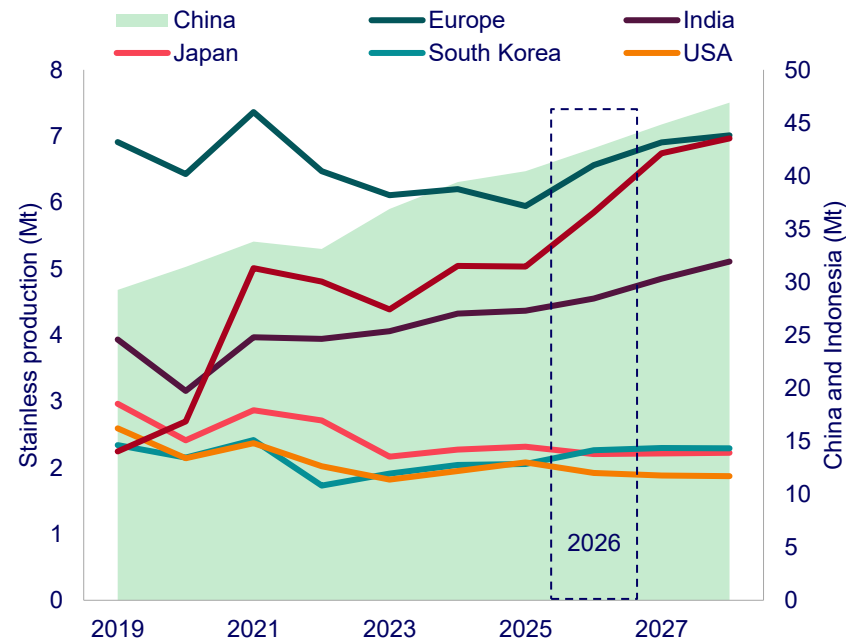
Source: Wood Mackenzie

Short-term outlook for stainless steel

Stainless markets recover everywhere although for some this may prove optimistic

- Growth of 1.5% raised global stainless steel production to 63.7 Mt in 2025. A stronger China, new capacity in Indonesia, sustained growth in India and a hopeful rebound in Europe pending a decline in Asian imports, supports 5.7% growth in 2026, to 67.7 Mt. Growth continues at around 4.5% a year, to 74.2 Mt in 2028.
- Chinese mills exceeded guidance in December and plan to raise output in January by 4%. This equates to a 25% improvement on January 2025, which could be seen as a boost to nickel prices although this may ultimately only balance the seasonal decline in February due to the Spring Festival. This leaves growth at 2.6% in 2025, to 40.5 Mt, and 5.4% in 2026 to 42.6 Mt.
- In Indonesia, Delong's closure of the Obsidian mill extended through the whole of Q4 2025 and will extend through January. Yet growth in Indonesian stainless production will continue whatever happens due to Jindal's new capacity, which was due for completion by the end of 2025. Assuming Obsidian also returns we could see 16% growth in 2026, to 5.9 Mt.
- We estimate European stainless contracted for the third year in four, down 4% to 5.9 Mt in 2025. Given the low base it is reasonable to expect at least some growth in 2026. However, the support anticipated to come from CBAM and revised safeguard tariffs may take a while to have a positive impact, leaving our forecast 10% growth in production, to 6.6 Mt, in danger of a quick cut-back.

Stainless steel production outlook by country



Source: Wood Mackenzie

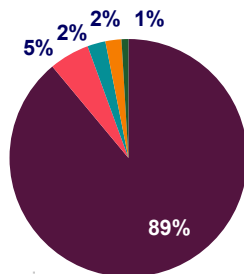
Short-term outlook for nickel in battery pCAMs

The long-lived positive outlook for nickel chemistries is no longer quite so positive

Global nickel consumption in pCAMs increased by 5% in 2025, to 543 kt, and this is forecast to rise by a further 10% in 2026, to 600 kt. This is supported by new capacity in China and Indonesia and may receive a boost from a surge in Chinese production to beat the 1 April cut in export rebates. As China adds and fills further capacity, growth will accelerate through to 2028, with global nickel consumption in pCAMs rising to 813 kt in that year.

Although it remains unclear how the growing popularity of LFP will undermine nickel ternary production rates, batteries for electrification, especially in electric vehicles, continues to be a growth story for nickel, just not the growth story it once was.

China accounts for 80% of global production, with Japan and South Korea making up most of the rest.

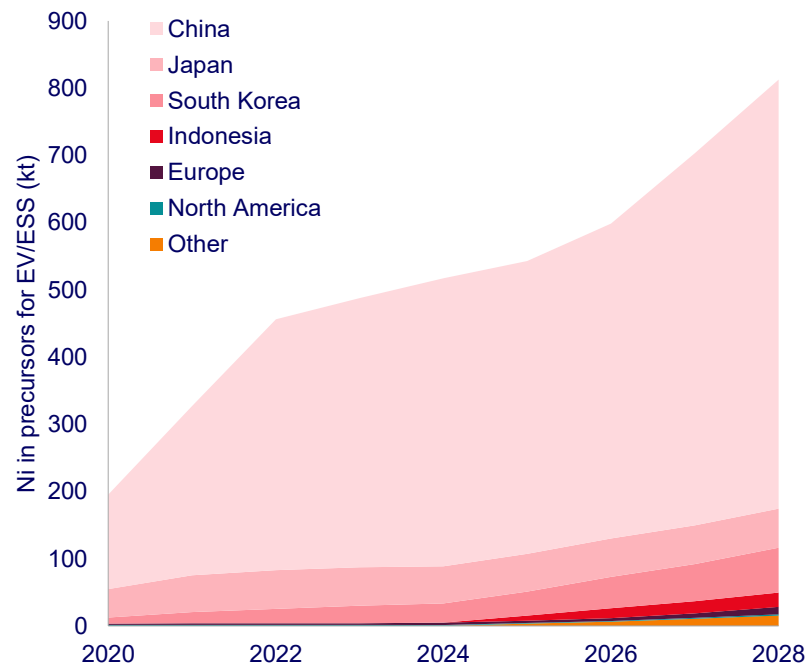


2026

- Electric vehicles
- Energy storage systems
- Portable electronics
- Power devices
- Motive products

Source: Wood Mackenzie

Global nickel consumption in pCAMs



Source: Wood Mackenzie

Vietnam to get its first stainless hot-rolling facility and, perhaps, a melt shop - 1

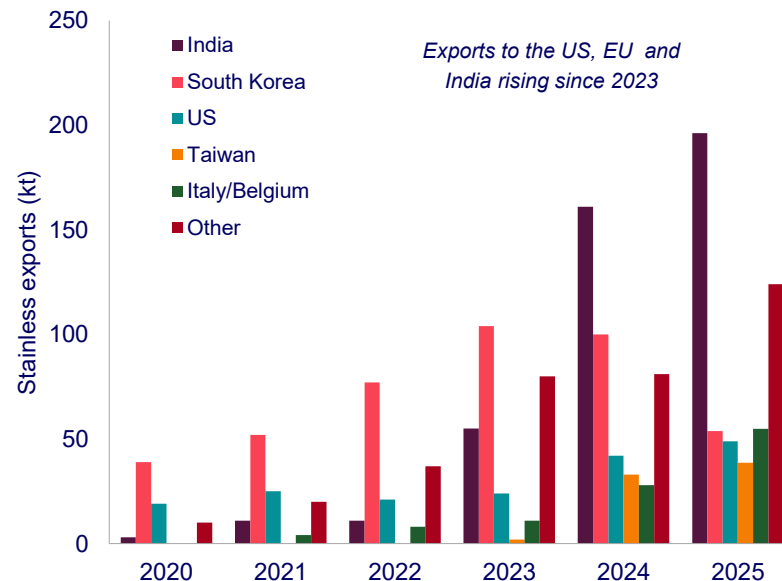
Yongjin Technology, a stable-mate of Tsingshan's, plans to integrate upstream from cold-rolling

Chinese stainless steel cold roller, Yongjin Technology Group, has announced plans to further expand its manufacturing footprint in Vietnam, reinforcing its strategy of offshore production amid tightening global trade barriers on Chinese-origin stainless. The proposed expansion in Ha Tinh province will complement Yongjin's existing cold-rolling (CR) stainless steel facility in Tien Giang Province, which has an estimated capacity of 500 ktpa and became operational in 2022.

Yongjin is planning a two-phase expansion with reported investment of US\$380 million currently named Singvin Alloy New Materials, with Yongjin holding 68.4% of the venture through its Singapore subsidiary Singvin Asset Management. Phase 1 will build a 2 Mtpa stainless hot-rolling (HR) mill while Phase 2 will add a melt shop of the same size producing stainless steel slab. The project is subject to phased execution and regulatory approvals and no timeline was stated. It is positioned as a "green, short-process" stainless facility, signalling an emphasis on lower-emissions, greater energy efficiency and downstream value addition.

This investment can be viewed as a strategic response to escalating trade protectionism faced by Chinese stainless steel exports. Over recent years, Chinese-origin flat stainless products have been subject to anti-dumping duties, safeguard measures, and import tariffs across the EU, the US, India, and parts of Southeast Asia, materially weakening China's export competitiveness. By relocating downstream processing to Vietnam, Yongjin has reaped the benefits of avoiding tariffs by qualifying products under non-Chinese rules of origin. This expansion might even leave Yongjin product immune to the EU's proposed "melt and pour" limitations.

Vietnam exports of cold-rolled stainless



Source: Global Trade Tracker; Wood Mackenzie
2025 annualised based on January-to November data

Vietnam to get its first stainless hot-rolling facility and, perhaps, a melt shop - 2

The new plant would mean imported semis from Tsingshan Indonesia would no longer be needed

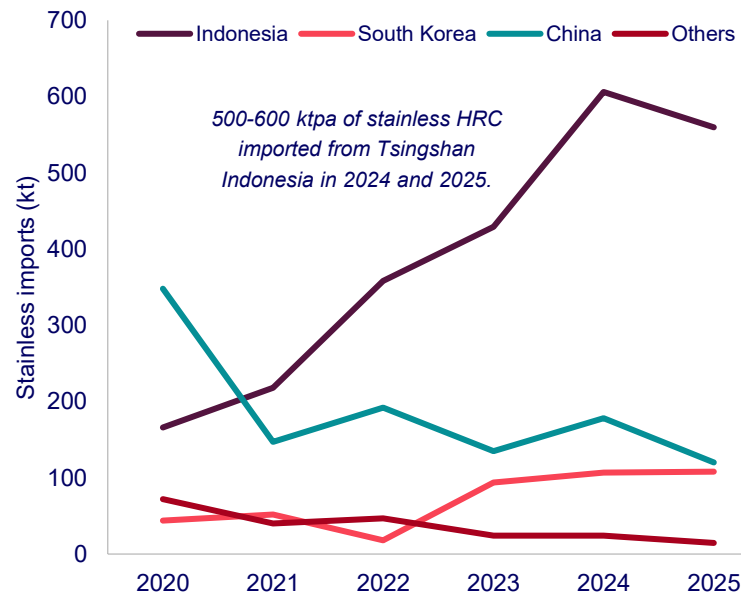
Yongjin's Vietnam strategy comes a few months after it released plans to expand its rolling operations to Turkey, where it intends to invest approximately US\$150–200 million to establish a 400 ktpa stainless CR mill. This is perhaps in response to EU circumvention duties on stainless imports from Turkey, much of which was of Chinese/Indonesian origin processed by Yongjin in Vietnam. The proposed Turkish facility, presumably, would avoid such a tariff.

Together, the Turkey and Vietnam investments illustrate a consistent pattern: offshoring capacity in such a way to avoid existing anti-Chinese protectionism while retaining Chinese ownership, technology, and commercial control.

Vertical integration with melting means that Yongjin would be self sufficient in stainless semis, having enough HR coil (HRC) to fill both its existing Vietnamese CR mill and the planned one in Turkey, and still have spare product to sell. However, this will also mean the 500-600 ktpa of HRC from Tsingshan in Indonesia that it has been dependent on since start up, will no longer be needed. Tsingshan will have to find another home for this or reduce output at its Indonesian plant.

Yongjin's plan to produce "greener" stainless will increase its product competitiveness abroad, such as in Europe where CBAM is now penalising emissions. Yongjin could become greener still if it pursues a scrap-based flow sheet in the melt shop, potentially importing scrap from other markets where scrap availability is already tight.

Vietnam imports of hot-rolled stainless



Source: Global Trade Tracker, Wood Mackenzie

European stainless

Market remains challenging as Italy offers subsidies to encourage “green” innovation

According to Aperam, the European stainless market has carried on from where it left off in December 2025. While the company remains happy with its prior forecast for Q4 2025 (due to be announced on 6 February) it also reaffirmed the challenging nature of the European market with its order book showing “no signs of recovery”. More significantly, Aperam also clarified that the positive impact of the EC’s proposed updated safeguards will not be felt until the second half of 2026, while the benefits of CBAM will likely take longer.

The problem with CBAM, as one British stockholder noted, is that the benchmark and default tariffs were released so late that companies planning to import stainless this year are having to order “blind” without knowing the full impact of those tariffs on the final price. This at least may turn some away from the import market to boost orders at the European mills.

The same company is seeing little happening in end-user markets: automotive and catering are slow, oil and gas is weak (reiterated by Aperam) with the only hope in the construction segment tied to data centres. Inventories stand at around three months and the knock-on effect of sharply rising nickel prices on the stainless prices has not triggered any “panic” buying. The attitude amongst distributors seems to be one of conserving cash and being well covered for whatever prices do.

While Aperam’s alloy specialty business has also been impacted by the weak oil and gas sector such value-add products remain key for the stainless makers, particularly as aerospace markets get stronger and defence spending increases. But as Outokumpu has demonstrated with its customers’ interest in green, low-carbon-emission stainless, this too falls in the value-add category.

This month, Kallanish noted that Italy’s 2026 budget plan includes incentives to produce “green” stainless steel determined by scrap inputs of 90% and energy consumption that remains below gradually tightening thresholds. The incentive amounts to subsidies for stainless companies of up to a total of €35 million (US\$41 million) a year until 2028.

Allegedly, Arvedi AST (Terni) is understood to be the main beneficiary and access to these funds will undoubtedly support the company’s ongoing investment programme in the facility and ease Terni’s way into what appears to be a growing niche of value-add “green” stainless products, one that more customers might turn to if CBAM progressively turns away high-carbon-emission stainless imports from Asia.

While greater scrap use would mean lower primary nickel uptake at Terni, the volumes of high-scrap stainless would have to be substantial to move Italy’s national scrap ratio significantly above our current estimate of 60%.

Nickel alloys

Aerospace and defence applications rising sharply

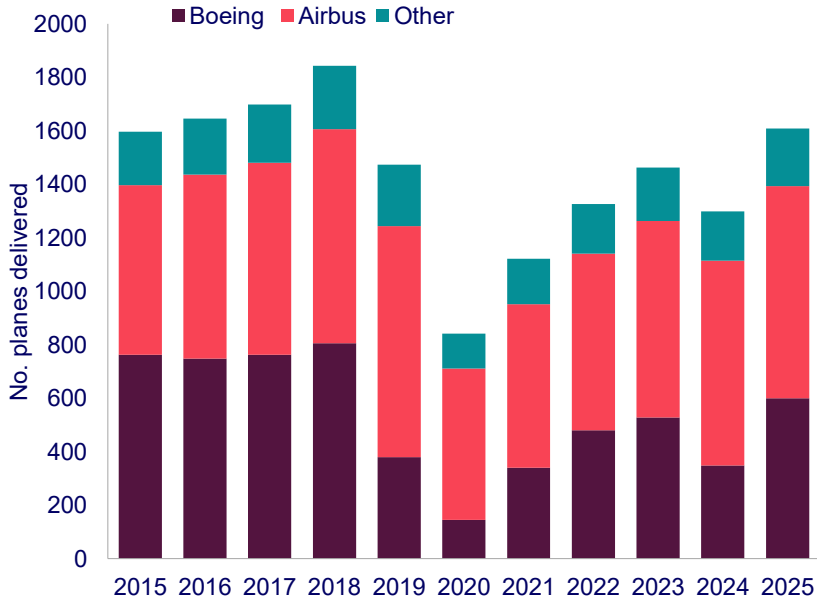
The challenges faced by Europe’s stainless mills have been partly mitigated by their expansion into higher value products, none more so than by the recent acquisitions of nickel alloy specialists, VDM and Haynes International by Acerinox. Aperam has always been in this sector through its Imphy subsidiary, and Outokumpu now makes nickel alloy powders and is looking to do more in this field at Avesta, in Sweden.

The markets using these alloys are doing well just now. Commercial aircraft production rates were at their best for a long time in 2025, with Airbus and Boeing both delivering their biggest annual totals (793 and 600, respectively) since 2018-2019. Orders last year and the ongoing backlog would suggest that these delivery rates will increase in the short term which will mean stronger demand for nickel alloys and stainless in aerospace applications.

Military/defence spending is also on the increase everywhere and this too will translate into resurgent demand for nickel alloys in these applications. Revenues at two of the US’s biggest aerospace and defence companies RTX and GE Aerospace increased by 30% and 27%, respectively, in 2025. And this can only improve given recent defence budgeting, such as €800 billion of potential defence spending tied to Readiness 2030 in the EU and the proposed US\$1.5 trillion budget in the US for fiscal 2026.

This could easily translate into growth in nickel demand in these applications that could exceed our forecast CAGR of 3% between now and 2030. Clearly this will boost global nickel demand but will do little on its own to counter the outlook of oversupply continuing until 2027.

Commercial aircraft deliveries



Source: Airbus; Boeing; Wood Mackenzie

Battery raw materials

EcoPro and CNGR start up new capacity

EcoPro BM Hungary

EcoPro BM completed its new campus in Debrecen, Hungary, and began commercial CAM operations in early January. The facility comprises two business entities: EcoPro Innovation, for cathode materials, CAM; and EcoPro AP for lithium processing. CAM capacity is 54 kt, all for high-nickel ternaries NCM and NCA, which might expand to 108 ktpa in the future.

At full capacity primary nickel consumption could reach of around 20-25 ktpa, which we expect will arrive as pCAM from EcoPro's joint venture with GEM in Morowali, Indonesia. If the company's portfolio expands to include non-nickel LFP, as it might in the future, nickel consumption will be lower.

CNGR Kalimantan, Indonesia

In Indonesia, CNGR completed construction on its new pCAM project in South Kalimantan and began production on 13 January 2026. Capacity of the plant is 20 ktpa of nickel ternary pCAMs that would require around 40 ktpa of nickel sulphate feed (approx. 9-10 ktpa Ni contained).

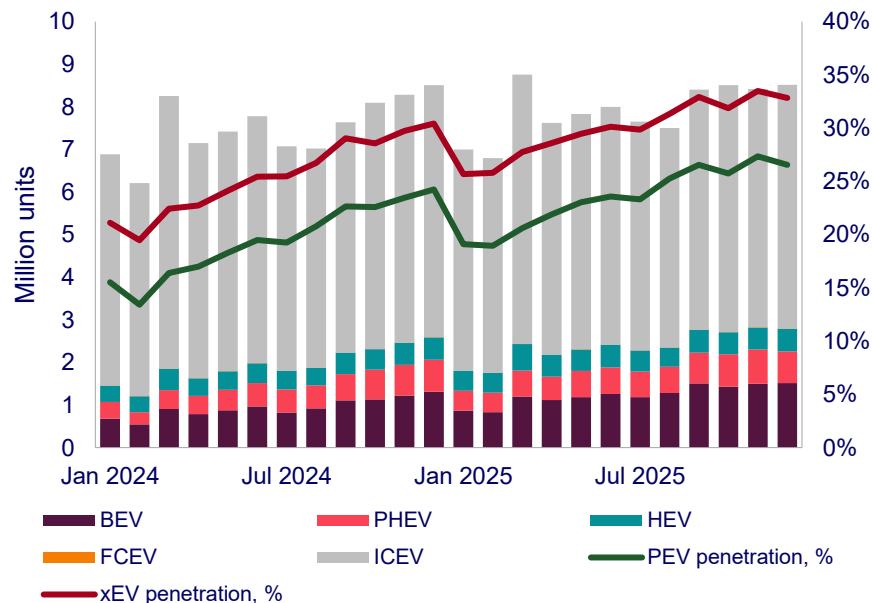
The plant aims to supply consumers outside China and start-up has been well timed to coincide with a reduction in the 13% VAT export rebate on pCAM leaving China, which starts on 1 April. With the cost of exports rising, this policy may accelerate the offshoring of some Chinese pCAM ventures to other countries, including Indonesia.



December sales remain flat on 2024, at 8.5 million; full-year sales up 5% at 95 million

December PEV penetration at 27%; full-year PEV penetration at 24% (up 4% from 2024)

Global vehicle sales and EV penetration



Source: Marklines, Wood Mackenzie. PEV = Battery-electric (BEV) + plug-in hybrid (PHEV), xEV = All powertrains except ICEV

China: The region registered 2.7 million passenger vehicle wholesales in December, dropping 7% month-on-month and 8% year-on-year.

New car sales in 2026 are expected to remain steady thanks to Beijing's recent decision to extend the vehicle scrappage/trade-in subsidy scheme. However some softness in growth is expected due to the changes to the scheme – the subsidy now applies only to cars priced above a certain threshold and the subsidy value has moved from a fixed amount to one based on the new car's price.

North America: The region registered 1.8 million vehicle sales in December, rising 12% on November but staying flat compared with 2024.

December saw an uptick in sales thanks to end-of-year discounts, rebates and low financing options from OEMs and dealers. 2025 saw US sales uplifted by the pull-forward effect of buyers advancing purchases in anticipation of upcoming tariff-driven price hikes. Whether the pull-forward effect will continue in 2026 will largely depend on whether OEMs, newly advantaged by eliminated penalties for non-compliance with federal fuel economy norms, choose to absorb tariff costs instead of passing them on.

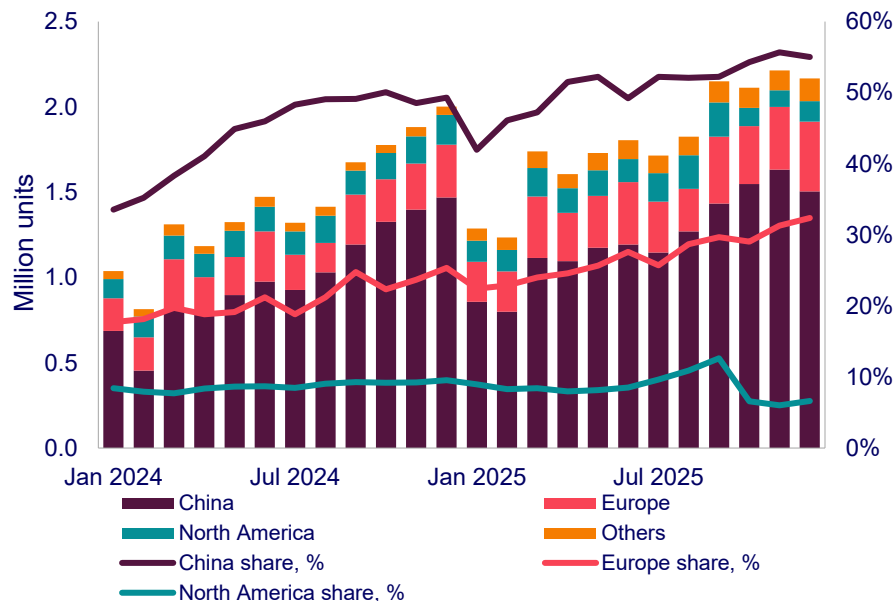
Europe: The region registered 1.27 million passenger vehicle sales in December, rising 8% month on month and 4% year on year.

December sales growth was largely driven by Germany, Italy and UK. 2026 new car sales growth will largely be driven by replacement cycles of corporate/fleet markets.

December PEV sales up 10% on 2024, at 2.16 million; full-year sales up 27% at 22.4 million

December PEV penetration at 55% for China, 32% for Europe and 7% for North America

PEV sales and penetration by region



Source: Marklines, Wood Mackenzie, PEV = Battery-electric (BEV) + plug-in hybrid (PHEV)

Note: Numbers in the chart specific to the Passenger segment for China and Europe

China: Passenger EV wholesales reached 1.5 million in December, dropping 8% on November and rising 2% on 2024. Monthly EV penetration dropped to 55%.

New auto pricing rules, EV energy consumption standards and the partially eliminated purchase tax exemption for EVs will marginally affect China's 2026 EV growth prospects. Volatile cell pricing – driven by rising lithium prices, measures to implement minimum-pricing within the battery midstream and reduction in the VAT export rebate – has the potential to impact pricing and sales of low-cost EVs.

Europe: Passenger EV sales hit 409,000 in December, rising 11% on November and 32% on last year. Monthly passenger EV penetration hit 32%.

The EU's decision to provide a minimum price undertaking option for Chinese EV importers looking to sidestep countervailing duties is expected to bias regional EV sales growth away from PHEVs (that were exempt from the added duties) back towards BEVs.

North America: EV sales touched 120,000 in December, rising 23% on November but dropping 31% on 2024. Monthly EV penetration improved slightly to 7%.

The US continued to experience the after-effects of Q3's "pull-forward" surge in EV sales prior to the expiry of the federal EV subsidy. EV sales in 2026 are expected to be muted due to the elimination of the penalties associated with the fuel economy norms combined with US OEMs' recent decisions to write off EV investments in favour of ICEVs and hybrids.

Supply

Refined supply

Refined output to increase by 10.7% by 2028 compared with 2025

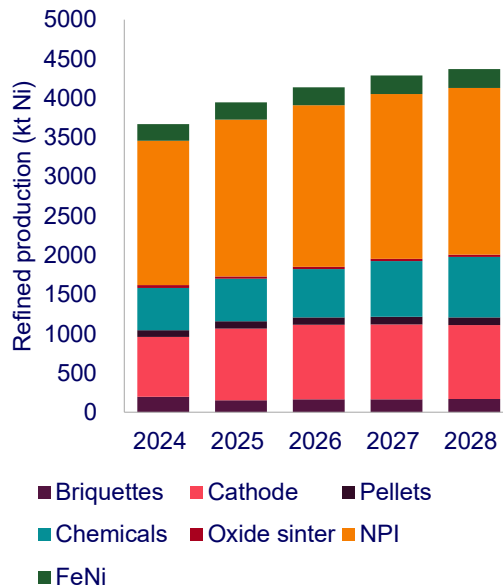
Class 1 output is forecast to increase by 4.3% by 2028, from 2025 levels, primarily from the refining of excess Chinese sulphate to metal.

Chemicals output will increase by 42% over same period. Most growth will occur in China, but important contributions will also come from South Korea and Indonesia.

Growth in FeNi is forecast to rise by 8% by 2028 mainly through new line addition and expansions in Indonesia. The likely loss of Tagaung Taung in Myanmar in terms of tonnes should be more than compensated for by the return to production at Pronico in Guatemala – see [slide #31](#).

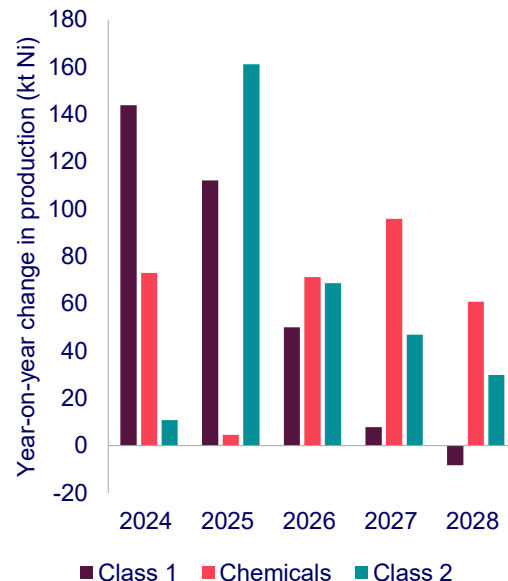
Meanwhile, NPI should increase by 6% with continued expansion in Indonesia at PT Gunbuster and Halmahera Jaya.

Refined production by type



Source : Wood Mackenzie

Growth in refined production by class



Source : Wood Mackenzie

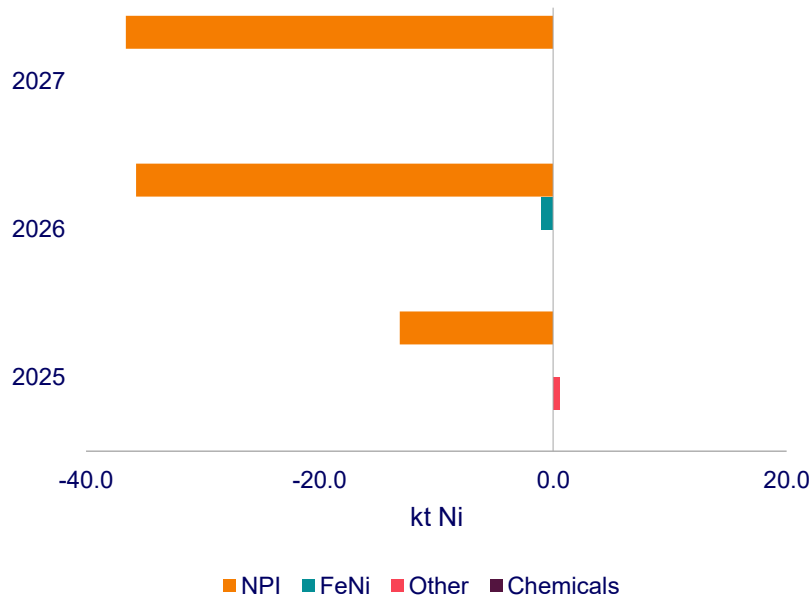
Changes in forecast nickel production

Downward adjustment in NPI to avoid double counting

Following on from our re-calculation last month of NPI being shipped from Indonesia to China and converted to matte by Jinchuan, we have made a similar revision to account for conversion of NPI to sulphate by Brunp. As noted in our December STO, Brunp has been using up to 12 kt a month (gross) of NPI from its Indonesian smelter and converting that to nickel sulphate at Yichang in China.

As with the Jinchuan product switch, we believe that Brunp is not alone and that conversion might be more widespread.

Main changes in finished nickel output



Source: Wood Mackenzie

Matte and MHP payability forecasts

Intermediate payabilities expected to stay high from 2026 onwards

Both matte and MHP payabilities have climbed steadily through to their current respective levels of 92% and 90%, hovering near historical highs. The payabilities are expected to remain elevated in the near term, driven by robust demand for the cobalt co-product and stockpiling demand from battery sector. The DRC's new cobalt export quota system has caused supply tightness and further underpinned the high payability levels.

In addition, sulphur prices surged sharply in 2025 and extended the rally into early 2026 amid disrupted Russian supply, which significantly raised MHP production costs.

	2025	2026	2027	2028
Matte	88%	91%	92%	92%
MHP	86%	89%	90%	90%

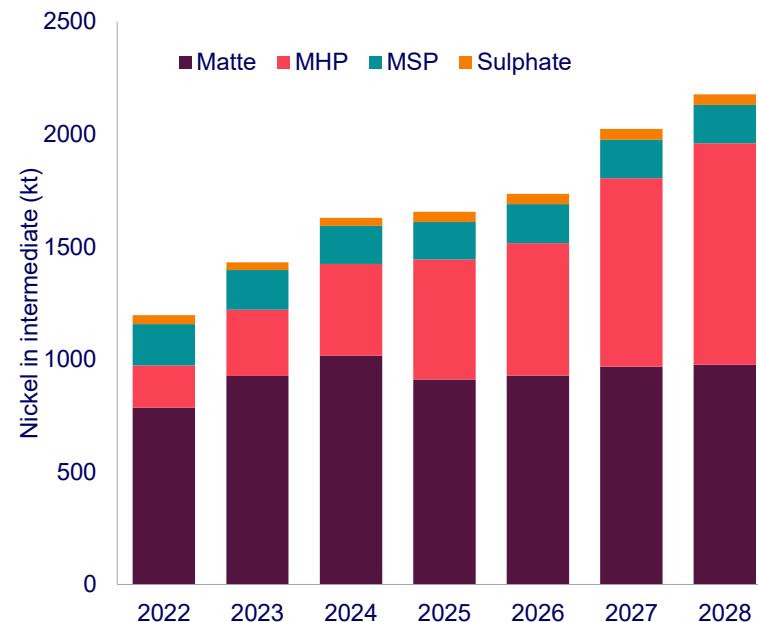
Source: Wood Mackenzie

After decreasing in 2025 to 911 kt, nickel-in-matte production will increase by 65 kt, or 7% by 2028.

Nickel-in-MHP output will increase by 85% over the same time frame, to 985 kt, primarily through capacity additions in Indonesia.

There remains considerable potential for the output of both MHP and matte to increase more strongly than shown, almost entirely from increases in Indonesian production.

Nickel-in-intermediates production



Source: Wood Mackenzie

Nickel sulphate production

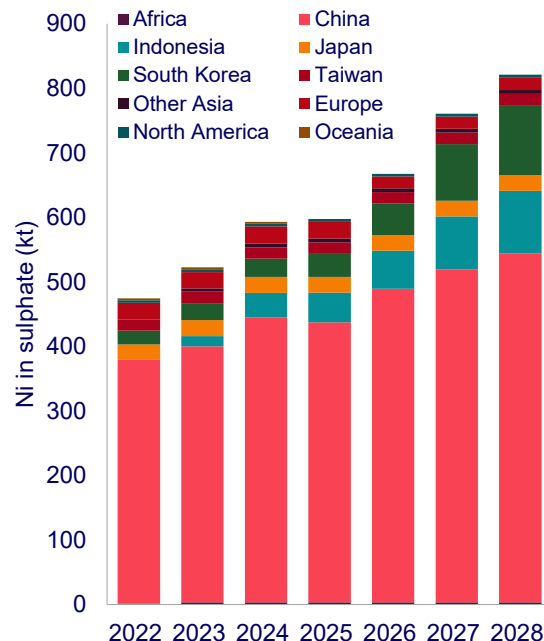
Nickel-in-chemicals production (predominantly as sulphate) will increase by 12% a year through to 2028

Last year China accounted for 73% of global nickel sulphate production, or 434 kt contained nickel. Increases at Brunp, CNGR, GEM and Quzhou Huayou will raise the volume by 23% through to 2028, to 540 kt, but the country's proportion of global output will decline to 66% by that year owing to growth in Indonesia and South Korea. Korea will see two new plants start up, one this year (Ulsan) and one in 2027 (Onsan).

The commissioning of two new sulphate plants in Indonesia in 2025 has added 80 ktpa nickel-in-sulphate capacity (QMB, 30 ktpa, and Huaxiang, 50 ktpa).

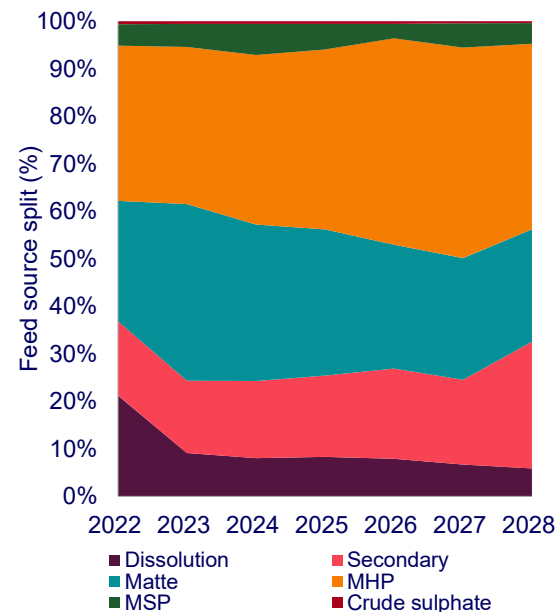
With increasing usage of MHP in Indonesia for sulphate (and metal) production, MHP availability has tightened somewhat in China. This suggests that pCAM producers like Huayou, Zhongwei and GEM (all of which have metal and sulphate production capabilities as well) will focus on sulphate output in the near term.

Global nickel sulphate production



Source: Wood Mackenzie

Raw material feeds for NiSO₄



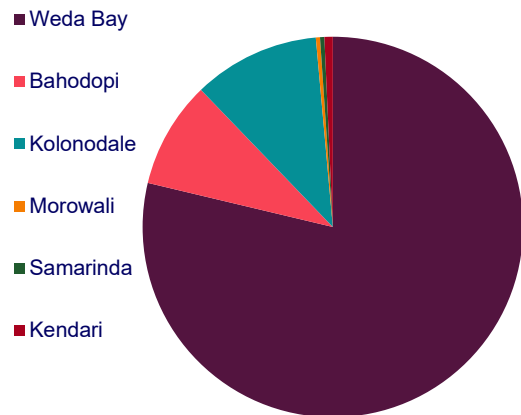
Source: Wood Mackenzie

Ore from the Philippines to Indonesia – explosive growth, significant potential

Filipino miners do not want to process ores at home and will be content to export ores

Indonesian ore imports by region*

2025: 15.4M GWT



One of the more dramatic dynamics taking place in Asia over the past two years has been the rise of ore exports from the Philippines to Indonesia. Miners in the Philippines had exported 30-40 Mtpa ore to China, Japan and South Korea. But once the Indonesian government began talking about imposing restrictions on miners at home, Philippines miners sensed an opportunity.

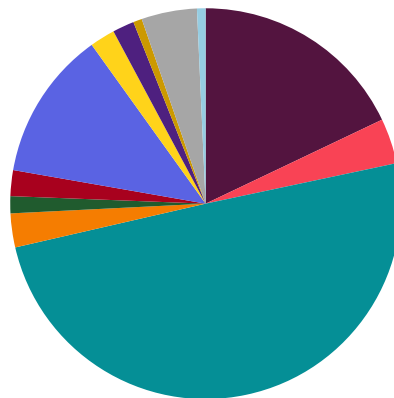
Starting late-2023 with a handful of shipments only, exports surged to 9.5 Mwmt* in 2024, and rose again last year to over 15 Mwmt, containing almost 135 kt nickel in ore.

Most of the ore came from three regions within the Philippines, and almost 70% of it was shipped to Weda Bay. Interestingly, as much ore was shipped to Indonesia during Q4 as in Q2, turning over the long-held belief that monsoon seasonality held up such trade. This reflects the development of new mines in zones which do not suffer as much in the wet season, which should change our expectations going forward.

*Mwmt – million wet metric tonnes

Filipino exports by district - 2025

- Dinagat
- Zambales
- Surigao
- Tawi-Tawi
- Isabela
- Homonhon
- Davao
- Manicani
- Palawan
- Nonoc
- Tubay
- Zamboanga



Source: Wood Mackenzie, Ferroalloy.net

Source: Wood Mackenzie, Ferroalloy.net.
* Ex-Philippines only

Nickel supply analysis – Myanmar's FeNi apparently offline, while Guatemala's seeks restart

Tagaung Taung struggling in Myanmar: Since 2022, production and export of FeNi from Tagaung Taung in Myanmar to China have been at best sporadic as the armed conflict between government forces and the Mandalay People's Defence Force flared up but then calmed down. In early 2025 we were encouraged by reports of improving relations between the warring parties that production had resumed in the area, and we noted that some FeNi was exported to China in May. Consequently, we reactivated its presence in our data and forecasts. However, May's trade appears to have been the sole activity for the year. At best, output has been erratic, while exports have remained undocumented. We will make additional enquiries but the probability is that our 2025 production estimate will be cut from 18 kt nickel-in-FeNi to 1-2 kt, while the 20 kt currently forecast for 2026 could be removed entirely. Tagaung's main customer has been Tisco Stainless Steel, and the lack of Tagaung's FeNi could push Tisco into taking western FeNi. Brazil, Colombia, South Korea and New Caledonia remain China's most significant suppliers of Class II products outside Indonesia.

Pronico re-enters the FeNi fray: Potentially offsetting any loss of FeNi this year from Tagaung Taung's misfortunes could be Solway Corp's Pronico smelter in Guatemala. A dispute between Solway's local subsidiary, Cia Guatemalteca de Niquel, and the country's Ministry of Mines forced the closure of Pronico and the Fenix mine back in 2022. But the two sides have resolved their differences, and the smelter is currently being reactivated. Given a planned April commercial start, we anticipate 2026 output of approximately 12 kt nickel-in-Feni, returning on an annualised basis to between 20-24 ktpa thereafter. Maximum historical output was in 2020 at 22.9 kt. The Fenix mine previously feeding the smelter is expected to be the main point of supply once again as the company is not yet preparing the second resource at Montufar for development. Consequently, we do not expect that ore in excess of the smelter's requirements will be made available for the export market. As this re-start confirmation came too late to be incorporated into this month's outlook, we will apply the necessary changes in February's report.

Nickel supply analysis – Brazilian Nickel sets up third non-binding MOU with potential offtakers

A rare example of progress outside Indonesia – but tight time frames for both production and customers

Non-binding MoUs

Brazilian Nickel (BRN)
Piauí – 99 Mt reserves
@ 0.84% Ni, 0.05% Co.

Average 28 ktpa Ni,
1 ktpa Co in MHP production
rate over 10 years using
heap leach.
Pilot plant built, exported trial
products in 2022-2023.
Several licences have been
obtained. BRN expects to
raise US\$1.45B total
financing by H1 2026.

20 January 2026:
USA: Westwin Elements
10 ktpa Ni, 240-400 tpa Co

25 March 2025
France: Electro Materials
Mobility Europe
10 ktpa Ni, 385 tpa Co

11 March 2025
Germany: Königswarter & Ebell
(Pure Battery Tech.)
5 ktpa Ni, 120-200 tpa Co

Proposed 18 ktpa Ni refinery using carbonyl
process. No decision yet on location but OK or
GA locations reviewed.

Public enquiry process on the plant near
Bordeaux to start “early 2026”. Approvals,
construction to be completed by H2 2027.
Output Ni+Co sulphates to ramp up in 2028.

Existing Ni+Co chemicals plant at Hagen has
been converted from 2.5 ktpa Ni-in-products to
13 ktpa total metals in pCAMs this year, using
black mass from recycler Cylib (at Dormagen)
and MHP.

Nickel supply analysis – NOAA cuts red tape for potential deep-sea miners

Operating outside the ISA, the US streamlines the bureaucratic process for mining the sea floor – TMC jumps in

USA and Japan: This month the US's Dept. of Commerce's National Oceanic and Atmospheric Association (NOAA) approved the consolidation of the prevailing two-step process for deep sea exploration and recovery into a single step, thereby hoping to promote activity within US waters. The move ties in with the Executive Order of 24 April 2025 from the White House entitled "*Unleashing America's Offshore Critical Minerals and Resources*". The enterprise at the forefront of mineral recovery from the sea floor, The Metals Company (TMC), wasted no time in taking advantage of the move by filing an application with NOAA to expand its hitherto range of exploration and recovery permits within the Clarion Clipperton Zone off Hawaii from 25,000 sq km to 65,000 sq km. This would increase its estimated resource to 619 Mt of sea floor nodules and offer an upside potential of an additional 200 Mt.

The merged permit ruling is significant for the timing of TMC's nodule processing plans at Pacific Metals Co's (Pamco) Hachinohe FeNi plant in Japan. Pamco has toll processed 3 kt of sea floor nodules through a modified FeNi furnace to make a matte which TMC has upgraded elsewhere into metallic products, such as Ni/Co/Mn sulphates and NiCuCo alloys. NOAA's reduced administrative process should allow TMC to scale up its nodules recovery and make more material available to Pamco by fiscal year 2028. By then, Pamco is expecting to move away from its conventional FeNi business, which has been hampered by poor ore feed availability and intense sales competition from NPI in what were its largest overseas markets (China and India).

Currently we do not know where the matte from Pamco/TMC would be upgraded at proposed commercial levels of 20-30 ktpa nickel content. TMC is heavily playing the "America First" card, yet no facility is likely to be built within the US by the known time frame. NOAA's expedited ruling will also encourage other US-based companies with seabed extraction activities to step up their investments. San Jose, CA-based Impossible Metals joined up with lead and energy recycler Aqua Metals last September, for example, to identify areas within US waters for extraction and processing.

Appendix

Key forecasts

Nickel key forecasts

	2025		2026		2027		2027	
Global (kt)								
Mine production	4215	+10.4%	4278	+1.5%	4659	+8.9%	4853	+4.2%
Smelter production	3916	+12.2%	3989	+1.9%	4319	+8.3%	4499	+4.2%
Refined production	3948	+14.7%	4055	+2.7%	4203	+3.6%	4360	+3.7%
Consumption	3525	+9.1%	3820	+8.4%	4098	+7.3%	4319	+5.4%
Market balance	423		236		104		41	
Cash LME price								
(US\$/t - current)	15154		15849		14844		16112	
(c/lb - current)	687		719		673		731	

Source: Wood Mackenzie

Nickel sulphate key forecasts

	2025		2026		2027		2028	
Global (kt Ni)								
NiSO4 production	597	+0.7%	668	+11.8%	760	+13.9%	821	+8.0%
Consumption in EV	543	+6.0%	599	+10.3%	703	+17.5%	813	+15.5%
Consumption in plating	40	+4.2%	41	+2.3%	42	+2.2%	43	+2.1%
Consumption in other	18	+9.4%	18	+2.5%	18	+1.3%	19	+1.3%
Total NiSO4 consumption	601	+6.0%	658	+9.5%	764	+16.1%	874	+14.4%
Market balance	-3		10		-3		-53	
NiSO4 price CHINA								
(US\$/t - current)	3353		3577	+6.7%	3419	-4.4%	3671	+7.4%
(US\$/t Ni - current)	15104		16113	+6.7%	15400	-4.4%	16537	+7.4%
Premium/discount to LME								
(US\$/t Ni = current)	-50		264	-628.9%	555	+110.5%	425	-23.4%
NiSO4 price ROW								
(US\$/t - current)	3756		3931	+4.7%	3719	-5.4%	4008	+7.8%
(US\$/t Ni - current)	16918		17705	+4.7%	16750	-5.4%	18055	+7.8%
Premium/discount to LME								
(US\$/t Ni = current)	1764		1856	+5.3%	1906	+2.7%	1943	+2.0%

Source: Wood Mackenzie; the Argus Media group

Quarterly nickel consumption (kt)

Quarterly global nickel consumption in stainless steel

	2025 Year	2026					2027				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
SS output	63731	15970	17295	17320	17108	67693	17054	18280	18238	17983	71555
Aust output	51149	12839	13971	14017	13896	54722	13782	14784	14767	14605	57938
Scrap (%)	33	33	33	31	31	32	32	32	31	32	32
Aust (%)	80	80	81	81	81	81	81	81	81	81	81
Primary Ni	2204	562	608	621	622	2413	613	652	655	647	2566

Source: Wood Mackenzie

Quarterly refined nickel consumption by first use

	2025 Year	2026					2027				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
TOTAL	3525	900	956	979	984	3820	986	1033	1040	1040	4098
Stainless steel	2204	562	608	621	622	2413	613	652	655	647	2566
Non-stainless	1321	338	348	358	363	1407	373	381	385	393	1532
Alloy steel	107	28	28	27	28	111	28	29	28	29	114
Non-ferrous alloys	321	83	84	85	86	337	86	87	87	88	348
Plating	182	46	46	46	46	186	47	47	47	48	190
Battery precursors	543	137	146	157	158	599	166	174	179	184	703
Foundry	80	21	21	21	21	84	21	21	21	21	85
Other	89	23	23	23	23	91	23	23	23	23	92

Source: Wood Mackenzie

Nickel stocks and premiums

	2025	2025 Q3	Q4	2025 Aug	Sep	Oct	Nov	Dec	2026 Jan
Stocks - kt									
Antwerp	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Baltimore	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Busan	70.0	55.7	70.0	51.2	55.7	66.2	68.0	70.0	73.2
Detroit	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Dubai	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Gwangyang	14.4	12.7	14.4	12.4	12.7	12.7	12.7	14.4	15.6
Johor	2.9	3.4	2.9	3.6	3.4	3.2	3.2	2.9	2.9
Kaohsiung	45.3	54.6	45.3	45.0	54.6	56.9	49.7	45.3	69.4
Liverpool	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mobile	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Port Klang	7.6	7.8	7.6	8.4	7.8	7.8	7.8	7.6	7.6
Rotterdam	33.9	31.8	33.9	31.2	31.8	32.4	33.0	33.9	34.8
Singapore (City)	78.3	63.1	78.3	55.7	63.1	71.1	77.3	78.3	80.0
Wissingen	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hong Kong	0.9	0.4	0.9	0.4	0.4	0.5	0.6	0.9	0.9
Other	0.0	-0.4	0.0	0.0	-0.4	0.0	0.0	0.0	0.0
Total LME stocks	255.3	231.1	255.3	209.8	231.1	252.8	254.4	255.3	286.3
Cancelled warrants	12.4	6.8	12.4	8.0	6.8	6.4	9.9	12.4	10.7

Shanghai Futures Exchange	45.5	29.5	45.5	26.4	29.5	36.8	40.8	45.5	50.8
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Prices									
LME									
Cash (US\$/t)	15155	15009	14888	14904	15098	15076	14686	14875	17807
3-month (US\$/t)	15355	15201	15083	15103	15281	15261	14888	15073	17995
Spread (3-month, cash)	200	192	195	199	184	185	202	198	188

	2025	2025 Q3	Q4	2025 Aug	Sep	Oct	Nov	Dec	2026 Jan
SHFE									
Front month settlement RMB/t	123380	121190	119325	120775	122460	122290	119640	117831	141320
3-month settlement RMB/t	123885	121487	119813	121074	122990	122917	120196	118426	141813
Front month settlement US\$/t	15181	14984	14900	14897	15211	15201	14888	14804	17929
3-month settlement US\$/t	15243	15021	14961	14934	15277	15278	14957	14879	17991
Physical premiums (US\$/t)									
Europe plating	324	322	302	331	304	281	277	349	475
Europe melting	309	334	269	318	320	280	272	256	413
Europe FPC	170	162	151	169	141	129	128	196	328
Exchange Rates/US\$									
Australian \$	1.554	1.528	1.526	1.540	1.516	1.528	1.538	1.510	1.490
Canadian \$	1.400	1.377	1.395	1.378	1.383	1.399	1.406	1.380	1.385
Yen	149.52	147.46	153.93	147.67	147.97	151.28	155.00	155.89	157.71
Yuan	7.192	7.158	7.087	7.175	7.124	7.120	7.112	7.044	6.975
Euro	0.754	0.856	0.859	0.860	0.852	0.860	0.865	0.854	0.856

Source: LME, SHFE, IMF, the Argus Media Group, Wood Mackenzie

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